MIG: Momentum Integrated Gradients

Transfer Attack on Face Verification

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Selecting MIG

Why MIG?

- Recent ICCV 2023 publication
- Official implementation available in TransferAttack repository
- Present in TransferAttack framework
- **Not implemented in intern baseline**

Code

```
https://github.com/Abhichy18/transferattacktInterns.git
```

Understanding MIG

MIG = Momentum Integrated Gradients targets Transferable Adversarial Attacks

Integrated Gradients (IG)

- Approximates path integral across baseline to input
- Averages gradients over S=20 interpolated steps
- Attributes key decision features for transferability

```
i_grad = (adv - x_base) * grad / S
```

Momentum Update

- Like MI-FGSM, maintains accumulated gradient
- g = decay * g + grad_norm
- adv = adv + alpha * sign(g)

Path Interpolation

- Generates S scaled brightness images
- Prevents overfitting to surrogate model
- Improves transferability across models

Attack Objectives (unchanged)

- Impersonation: maximize cosine similarity
- Dodging: minimize cosine similarity

Adapting MIG to Face Verification

Problem

- Official MIG tested on classification networks
- Baseline uses CNN-based DeepFace models:
  - ArcFace
  - Facenet512
  - GhostFaceNet
  - VGG-Face
- Direct copy-paste was impossible

Adapted Core Ideas

- Batch-level linear path interpolation
- Embedding cosine similarity loss mapping
- Integrated gradient normalization
- Momentum accumulation
- Epsilon-ball projection
- Valid pixel range clipping [-1.0, 1.0]

Code Integration

Step 1 — Register MIG

```
ALL_ATTACKS = [
  'PGD', 'MI_FGSM',
  'TI_FGSM', 'SI_NI_FGSM',
  'MI_ADMIX_DI_TI',
  'MIG' # <-- added
]
```

Step 3 — Dispatcher

```
if attack_name == "MIG":
    return mig_attack(...)
```

Step 2 — mig_attack() Function

1 Initialize adversarial image

2 Create S interpolated image copies

3 Batch forward pass through surrogate model

4 Compute embedding similarity & loss

5 Compute gradient via GradientTape

6 Calculate integrated gradient (i_grad)

7 Normalize integrated gradient

8 Update momentum

9 Project into epsilon-ball

10 Clip image to [-1.0, 1.0]

11 Repeat for NUM_ITER

Key Challenges Encountered

Missing Evaluation Pipeline

- No script existed to compute similarity scores
- Only the summarization script was present
- Evaluation pipeline had been intentionally removed
- Had to reverse-engineer from baseline CSV structure

Dataset Path Mismatch

- Professor's CSV used /content/face_module/...
- My Colab used dataset/...
- Solution: path conversion function

Model Input Size Mismatch

- ArcFace, GhostFaceNet: 112 x 112
- Facenet512: 160 x 160
- VGG-Face: 224 x 224
- Fixed per-model resizing

Reconstructed Evaluation Pipeline

Pipeline Flow

1 Load adversarial image

2 Load victim model

3 Compute embedding

4 Load target image

5 Compute target embedding

6 Compute cosine similarity

7 Store CSV row

CSV Schema

```
row_id
attacker_model
img1
img2
dataset
attack_type
victim_model
attack_method
variant
similarity
```

Output Files

- MIG_raw_similarities_long.csv
- Merged into combined_raw_similarities_long.csv
- Processed by build_subset_baselines.py

Final Results

Attack	Breach Rate
SI-NI-FGSM	31.46%
MIG (Proposed)	**34.17%**
MI-FGSM	26.04%
MI-ADMIX-DI-TI	23.96%
TI-FGSM	20.00%
PGD	16.46%

MIG outperformed:

- SI-NI-FGSM (+2.71%)
- MI-FGSM (+8.13%)
- PGD (+17.71%)

Why higher performance?

- Integrated gradients across 20 path steps
- Prevents overfitting to surrogate features
- Captures salient face representation paths
- Highly effective on CNN backbones

What I Learned

Technical

- Transfer attack mechanics
- Face verification pipelines
- Reading and adapting research code
- Reverse engineering incomplete repositories
- Building missing evaluation infrastructure

Process

- Integrating new research into existing codebases
- Quantitative attack comparison
- Debugging silent failures (missing rows, path mismatches)
- Extending baselines without redesigning the system

Thank You

Abhishek Choudhary — MSIT

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